American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Intermediate Division

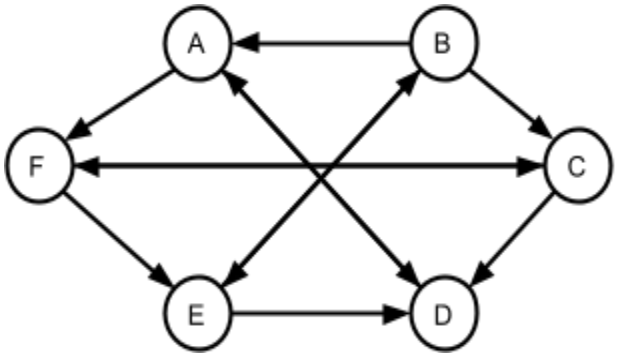
1. Graph Theory

Given a directed graph with vertices {A, B, C, D} and edges {AB, CB, AD, BC, BD, AC}, if edge DC were added, how many more paths of length 2 would there be?

A. 2
B. 3
C. 5
D. 6
E. 9

2. Graph Theory

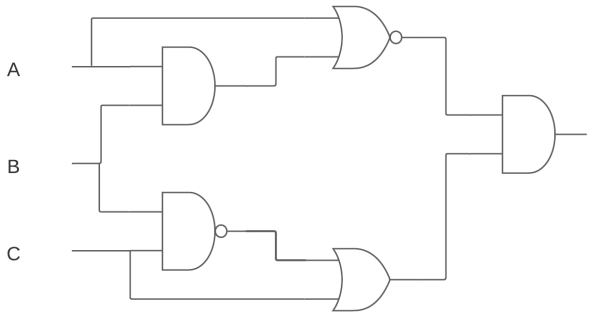
How many cycles of length 4 are there from A?



A. 1
B. 2
C. 3
D. 4
E. 5

3. Digital Electronics

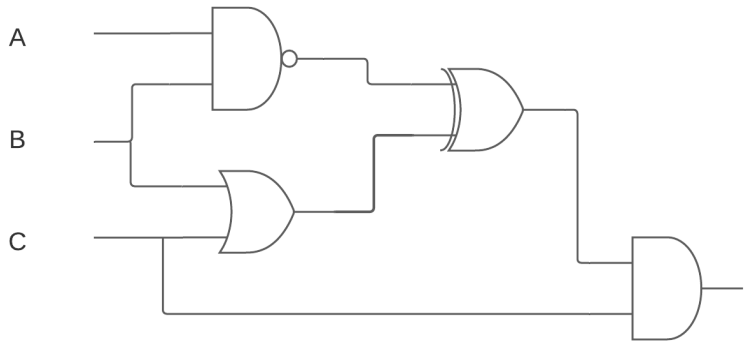
Simplify the Boolean expression represented by the following digital circuit.



A. $\overline{A}$
B. $\overline{A} + \overline{B}$
C. $\overline{B}$
D. $\overline{B} + \overline{C}$
E. $\overline{C}$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 0
B. 1
C. 4
D. 7
E. 8

5. Assembly Language

How many values are printed when this program is executed?

```
A           DC          12
B           DC          1
TOP         LOAD        A
            DIV         B
            STORE       C
            LOAD        B
            MULT        C
            STORE       D
            LOAD        A
            SUB         D
            BE          OUT
IN          LOAD        A
            SUB         B
            BE          STOP
            LOAD        B
            ADD         =1
            STORE       B
            BU          TOP
 OUT        PRINT       B
            BU          IN
 STOP       END
```

A. 2
B. 3
C. 4
D. 5
E. 6

American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Intermediate Division

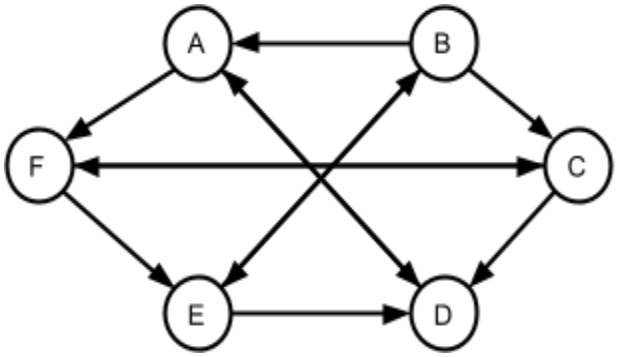
1. 图论

给定一个顶点为{A，B，C，D} 且边为{AB，CB，AD，BC，BD，AC}的有向图，如果加上边 DC，那么新增多少条长度为 2 的路径？

A. 2
B. 3
C. 5
D. 6
E. 9

2. 图论

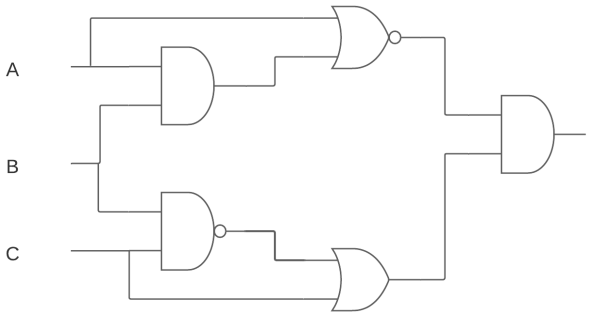
从 A 出发有多少个长度为 4 的环？



A. 1
B. 2
C. 3
D. 4
E. 5

3. 数字电路

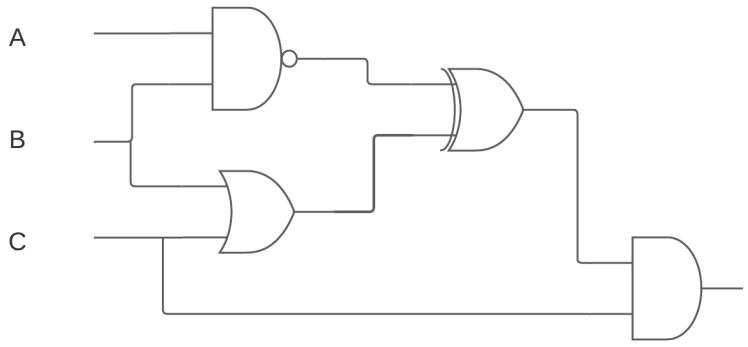
化简下述数字电路表示的布尔表达式。



A. $\overline{A}$
B. $\overline{A} + \overline{B}$
C. $\overline{B}$
D. $\overline{B} + \overline{C}$
E. $\overline{C}$

4. 数字电路

有多少个有序三元组能够使得下述数字电路为**真**？



A. 0
B. 1
C. 4
D. 7
E. 8

5. 汇编语言

执行以下程序**时**，会打印输出多少个值？

```
A           DC          12
B           DC          1
TOP         LOAD        A
            DIV         B
            STORE       C
            LOAD        B
            MULT        C
            STORE       D
            LOAD        A
            SUB         D
            BE          OUT
IN          LOAD        A
            SUB         B
            BE          STOP
            LOAD        B
            ADD         =1
            STORE       B
            BU          TOP
 OUT        PRINT       B
            BU          IN
 STOP       END
```

A. 2
B. 3
C. 4
D. 5
E. 6